
Generative and Agentic AI for Autonomous Network Operations: A Self-Healing Framework for Intelligent NOC Systems

Integrating Large Language Models and Agentic AI for Intelligent Incident Management in Telecom

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Abstract—Generative Artificial Intelligence (GenAI) is quietly transforming the way telecom networks are managed — not through incremental improvements, but by fundamentally rethinking how Network Operations Centers (NOCs) detect, analyze, and respond to problems. For years, NOC teams have been caught in a cycle of reactive monitoring and manual troubleshooting, where engineers scramble to identify issues only after something has already gone wrong. The result? Unnecessary downtime, stretched teams, and a Mean Time To Repair (MTTR) that no one is proud of. This white paper introduces a unified framework that brings together GenAI and Agentic AI to change that dynamic — moving telecom operations from firefighting mode to something far more intelligent and self-sustaining. At the heart of the framework are large language models (LLMs) that read through complex network logs, make sense of what is happening, and surface actionable recommendations in near real time. Alongside this, an agent-based architecture handles the follow-through, autonomously executing remediation tasks without constant manual oversight. Experimental evaluation demonstrates a 40–60% reduction in incident resolution time, alongside a meaningful drop in manual intervention, stronger service reliability, leaner operational costs, and a scalable architecture ready for next-generation autonomous telecom networks.

Index Terms—GenAI, Agentic AI, NOC Automation, Telecom, LLM, Root Cause Analysis, Self-Healing Networks, MTTR Reduction, Incident Management.

I. INTRODUCTION

Modern telecommunications infrastructure underpins virtually every facet of digital society—from consumer connectivity to enterprise-grade mission-critical services. As networks grow in complexity with 5G rollouts, multi-cloud environments, and edge computing deployments accelerating simultaneously, the operational demands placed on Network Operations Centers (NOCs) have reached an inflection point.

For years, NOC teams have been caught in a cycle of reactive monitoring and manual troubleshooting, where engineers scramble to identify issues only after something has already gone wrong. The result is unnecessary downtime, stretched teams, and a Mean Time To Repair (MTTR) that falls well short of modern SLA expectations. Traditional NOC operations that rely on rule-based monitoring systems, manual log triage, and reactive workflows are increasingly ill-suited to the scale, speed, and interdependency of contemporary telecom infrastructure.

This paper introduces a unified framework that brings together GenAI and Agentic AI to change that dynamic—moving telecom operations from firefighting mode to something far more intelligent and self-sustaining. Prodapt, as a leading telecom-focused technology and services company, commissioned this research to investigate whether these AI architectures could form the foundation of a genuinely

autonomous, self-healing NOC that detects, diagnoses, and resolves incidents with minimal human intervention.

A. Problem Statement

The core challenges of conventional NOC environments include:

- Alert fatigue from high false-positive rates (~40%) overwhelming on-call engineers
- Extended MTTR due to manual log analysis and escalation bottlenecks
- Siloed tooling preventing correlation across network domains
- Overreliance on tribal knowledge and institutional memory
- Inability to scale operational capacity linearly with network growth

II. BACKGROUND AND RELATED WORK

A. Evolution of NOC Automation

Network operations automation has progressed through several generations: from scripted runbooks and SNMP-based monitoring in the 1990s, through ML-assisted anomaly detection in the 2010s, to today's AI-native operations platforms [1]. Each generation reduced response time and improved reliability, but none achieved true end-to-end autonomy.

B. Large Language Models in IT Operations

The emergence of LLMs such as GPT-4, Claude, and open-source alternatives including Llama has opened new possibilities for natural language processing of operational data [2]. Unlike traditional NLP pipelines, LLMs possess contextual reasoning capability to interpret unstructured log data, correlate events across time windows, and generate human-readable root cause hypotheses.

C. Agentic AI Architectures

Agentic AI refers to systems in which AI models are endowed with tool use, memory, and the capacity to take sequences of actions toward a goal [3]. Frameworks such as LangChain, AutoGen, and CrewAI have demonstrated that multi-agent systems can handle complex, multi-step workflows autonomously. Applied to network operations, this paradigm enables closed-loop remediation where the AI not only diagnoses but also acts [4].

III. PROPOSED FRAMEWORK

The proposed GenAI-Agentic NOC framework is organised around four interconnected layers, each responsible for a distinct stage of the incident lifecycle. Together, they form a closed-loop system capable of transitioning from alert ingestion to autonomous resolution without requiring human touchpoints for the majority of incidents. Fig. 1 illustrates the layered architecture.

LAYER 4 — Feedback & Continuous Learning
↓ <i>Fine-tuning pipeline</i> <i>Outcome logging</i> <i>Model adaptation</i>
LAYER 3 — Agentic Execution Layer
↓ <i>Config Agent</i> <i>Restart Agent</i> <i>Rerouting Agent</i> <i>Escalation Agent</i>
LAYER 2 — LLM-Powered Analysis Engine
↓ <i>Event Correlation</i> <i>Root Cause Inference</i> <i>Remediation Synthesis</i>
LAYER 1 — Intelligent Data Ingestion
↓ <i>Logs</i> <i>SNMP Traps</i> <i>Syslogs</i> <i>KPIs</i> <i>Config Records</i> <i>Tickets</i>

Fig. 1. Four-layer GenAI-Agentic NOC architecture.

A. Layer 1 — Intelligent Data Ingestion

At the foundation, the framework aggregates structured and unstructured operational data from diverse sources: network element logs, SNMP traps, syslogs, performance management KPIs, configuration change records, and ticketing systems. A normalisation pipeline transforms heterogeneous data into a unified representation suitable for LLM consumption, supporting multi-vendor and legacy environments without bespoke integration work.

B. Layer 2 — LLM-Powered Analysis Engine

The analysis engine constitutes the cognitive core. An LLM fine-tuned on telecom-specific log taxonomies and incident resolution histories receives normalised event streams and performs three primary functions: (i) Event Correlation across distributed alerts; (ii) Root Cause Inference generating ranked probabilistic hypotheses; and (iii) Remediation

Synthesis producing structured task specifications for the execution layer.

C. Layer 3 — Agentic Execution Layer

Validated recommendations are passed to specialised sub-agents: a network configuration agent, a service restart agent, a traffic re-routing agent, and an escalation agent. Each sub-agent carries appropriate tool access, guardrails, and rollback capabilities. A supervisor agent maintains situational awareness across the full incident timeline [5].

D. Layer 4 — Feedback and Continuous Learning

Post-resolution, outcome data—including action effectiveness, resolution confirmation, and any human overrides—is fed back into the fine-tuning pipeline. This closed feedback loop enables continuous adaptation to evolving network topologies and emerging failure modes without requiring manual model intervention.

IV. TRADITIONAL NOC VS. GENAI-POWERED NOC

TABLE I summarises the operational delta across five critical dimensions between conventional NOC workflows and the proposed framework.

TABLE I
COMPARATIVE ANALYSIS: TRADITIONAL NOC VS. GENAI-POWERED NOC

Challenge Area	Traditional NOC	GenAI-Powered NOC
Incident Detection	Threshold alerts; ~40% false positives	AI anomaly detection; ~8% false positives
Root Cause Analysis	Manual log review; hours of effort	LLM inference; near-real-time
Remediation	Human runbooks; error-prone	Autonomous agent execution
MTTR	4–8 hours	1.6–4.8 hrs (40–60% reduction)
Scalability	Linear headcount growth	Elastic AI ops; minimal overhead

V. EXPERIMENTAL EVALUATION

A. Evaluation Methodology

The framework was evaluated against a synthetic dataset derived from anonymised incident records spanning twelve months of operations at a representative tier-1 telecom operator environment. The dataset comprised 4,200 distinct incident records across five network domains: RAN, core, transport, OSS/BSS, and cloud-native functions.

A baseline was established using the operator's existing rule-based monitoring and ticket management system. The

GenAI framework was then applied to the same incident corpus in a controlled simulation environment, with human engineers available for escalation but not for first-line intervention, ensuring a clean comparative measurement.

B. Key Performance Metrics

TABLE II presents the principal quantitative outcomes. The most significant improvement was observed in MTTR, which decreased from a baseline of 4–8 hours to 1.6–4.8 hours—a 40–60% reduction consistent across all five network domains evaluated.

TABLE II
PERFORMANCE METRICS: BASELINE VS. GENAI
FRAMEWORK

Metric	Baseline	GenAI Framework
MTTR	4–8 hrs	1.6–4.8 hrs
Manual Interventions / Incident	8–12	2–3
Escalation Rate	~35%	~12%
False Positive Alert Rate	~40%	~8%
Operational Cost Index	1.0x	0.55–0.70x

C. Qualitative Observations

Beyond quantitative metrics, the evaluation surfaced several important qualitative findings:

- Engineers reported significantly reduced cognitive load during high-severity incident windows
- LLM-generated root cause summaries were rated accurate or partially accurate in 89% of evaluated cases
- Agentic remediation actions required rollback in fewer than 4% of executions, demonstrating operational reliability
- The framework demonstrated graceful degradation, escalating appropriately when internal confidence thresholds were not met

VI. INDUSTRY USE CASES AT PRODAPT

A. 5G RAN Anomaly Detection and Self-Healing

5G Radio Access Networks generate enormous volumes of performance counters and KPI streams. The LLM analysis engine identified degraded cell behaviour up to 18 minutes before threshold breaches triggered conventional alarms, enabling pre-emptive parameter adjustments without service impact. This proactive capability is particularly valuable during peak demand periods and planned maintenance transitions.

B. Core Network Incident Triage

In packet core environments, cascading failures can propagate rapidly across domains. The agentic framework isolated the originating fault in multi-domain cascades and executed targeted isolation actions while protecting unaffected

traffic flows—a task previously requiring three-to-four specialist engineers working in parallel with significant coordination overhead.

C. OSS/BSS Process Automation

Order management and provisioning workflows within OSS/BSS systems frequently stall due to validation failures or upstream dependency issues. The framework's NLP capability interpreted error payloads from legacy systems without vendor-specific integration, reducing provisioning failure resolution time by approximately 55%.

VII. CHALLENGES AND CONSIDERATIONS

A. Hallucination and Trust

LLMs can generate plausible but incorrect root cause hypotheses—a risk that is unacceptable in mission-critical infrastructure. Production deployment requires confidence scoring, mandatory human confirmation gates for high-risk remediation actions, and robust audit trails for all agent-executed changes [6].

B. Data Privacy and Security

Network logs may contain sensitive subscriber or infrastructure information. The framework must be deployed within on-premise or private cloud boundaries, with data masking applied before any LLM inference call reaches an external service, particularly under GDPR and regional data residency regulations.

C. Change Management and Adoption

Autonomous remediation represents a significant cultural shift for NOC teams accustomed to manual control. Successful deployment requires a phased rollout—starting with AI-as-advisor mode before transitioning to AI-as-operator—alongside comprehensive retraining programmes that reposition engineers as oversight and exception handlers.

D. Integration Complexity

Telecom environments typically include legacy OSS/BSS systems, proprietary network element interfaces, and multi-vendor equipment spanning multiple generations. The ingestion layer must support diverse protocol adapters, and the agentic execution layer requires vendor-approved APIs to ensure autonomous actions comply with change management requirements.

VIII. ROADMAP AND FUTURE WORK

This research constitutes an early-stage proof of concept. The pathway to production deployment is structured in four phases:

- **Phase 1 (0–6 months):** Shadow mode integration; LLM outputs presented as recommendations only with no autonomous execution
- **Phase 2 (6–12 months):** Controlled autonomous remediation for low-risk, high-frequency incident categories with full audit logging

- **Phase 3 (12–24 months):** Full agentic autonomy with human oversight dashboard; continuous fine-tuning pipeline operational
- **Phase 4 (24+ months):** Cross-operator federated learning; integration with network digital twin for simulation-based validation prior to production execution

Future research directions include multimodal inputs incorporating topology diagrams, formal verification methods for agent action safety, and proactive capacity management through GenAI-driven traffic forecasting [7]. The application of reinforcement learning from human feedback specific to NOC operational contexts also warrants investigation.

IX. CONCLUSION

This paper has presented a unified framework that brings together GenAI and Agentic AI to move telecom operations from firefighting mode to something far more intelligent and self-sustaining. At its core, large language models read through complex network logs, surface actionable recommendations in near real time, and an agent-based architecture handles the follow-through—autonomously executing remediation tasks without constant manual oversight.

Experimental evaluation demonstrates a 40–60% reduction in incident resolution time, alongside a meaningful drop in manual intervention, stronger service reliability, leaner operational costs, and a scalable architecture ready for next-generation autonomous telecom networks. For Prodapt’s clients, this translates directly into reduced operational expenditure, improved SLA adherence, and the capacity to redeploy skilled engineers toward higher-value innovation work.

Prodapt is uniquely positioned to lead this transformation—combining deep telecom domain expertise with emerging AI engineering capability. This research demonstrates that the long-standing aspiration of self-healing, autonomous network operations is no longer a distant vision. With the right architecture, it is an achievable near-term reality.

ACKNOWLEDGMENT

The author thanks the Prodapt Network Automation Practice leadership for their guidance and domain expertise throughout this research initiative. This work was conducted as part of the Prodapt Intern Research Program, 2025.

REFERENCES

- [1] Y. Deng et al., "Towards autonomous network operations: A survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 3, pp. 1580–1612, 2023.
- [2] T. Brown et al., "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.
- [3] S. Yao et al., "ReAct: Synergizing reasoning and acting in language models," in *Proc. ICLR*, Kigali, Rwanda, May 2023.
- [4] A. Madaan et al., "Self-refine: Iterative refinement with self-feedback," in *Proc. NeurIPS*, New Orleans, USA, Dec. 2023.
- [5] M. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, virtual, Dec. 2020.
- [6] X. Wang et al., "Self-consistency improves chain of thought reasoning in language models," in *Proc. ICLR*, Kigali, Rwanda, May 2023.
- [7] ETSI, "Zero-touch network and service management (ZSM); reference architecture," ETSI GS ZSM 002, v1.1.1, Aug. 2019.
- [8] ITU-T, "Autonomous networks — architecture and framework," ITU-T Recommendation Y.3172, Jul. 2019.
- [9] S. Liu et al., "AIOps: Real-world challenges and research innovations," in *Proc. ICSE*, Seoul, South Korea, May 2020.
- [10] Prodapt Solutions, "Network automation practice: Internal research initiative," Prodapt Technical Report TR-2025-NAP-001, 2025.